One-dimensional PCA

Archer Yang

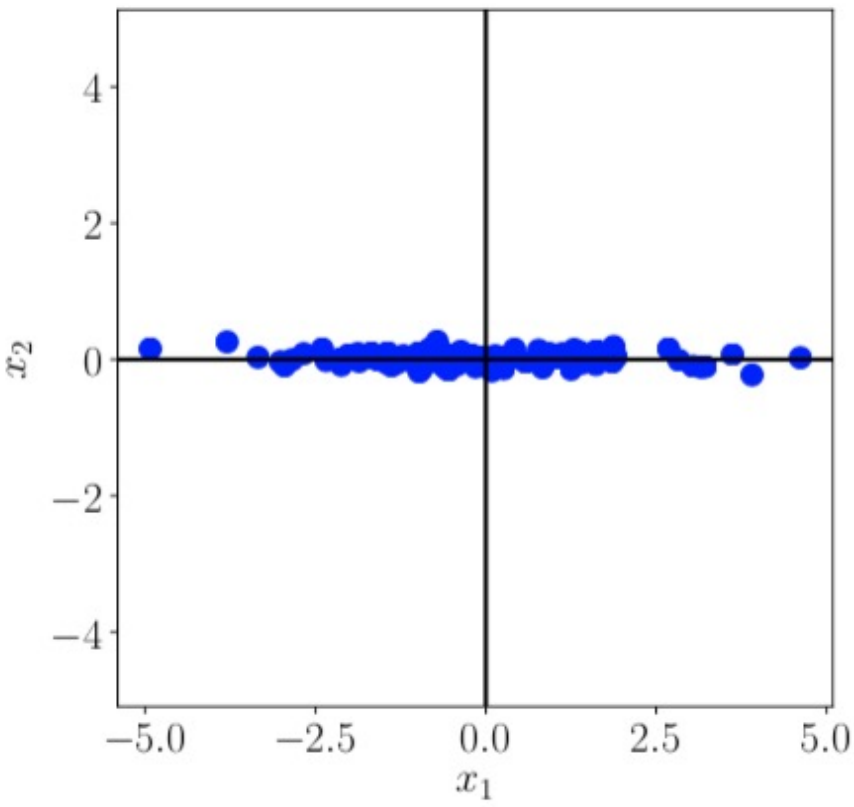
McGill University

- **Reading assignment:** read chapter 10 of Mathematics for Machine Learning book.

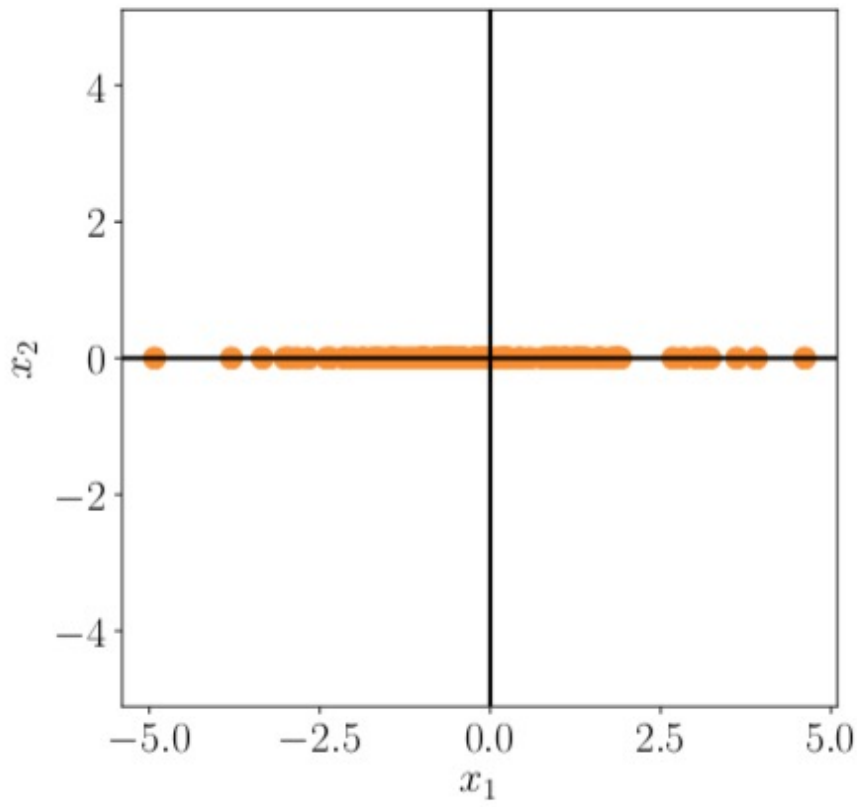
Dimension reduction

- High-dimensional data is very common in real-life applications, e.g. in genomics, finance, e-commerce etc.
- Dimensions in high-dimensional data are often **correlated** — it has an **embedded lower-dimensional structure**.
- Dimensionality reduction exploits such low dimensional structure
  - Compression
  - Visualization
  - Data generation
- We study **principal component analysis** (PCA) – linear dimensionality reduction by Pearson (1901) and Hotelling (1933).

One-dimensional PCA



(a) Dataset with x_1 and x_2 coordinates.



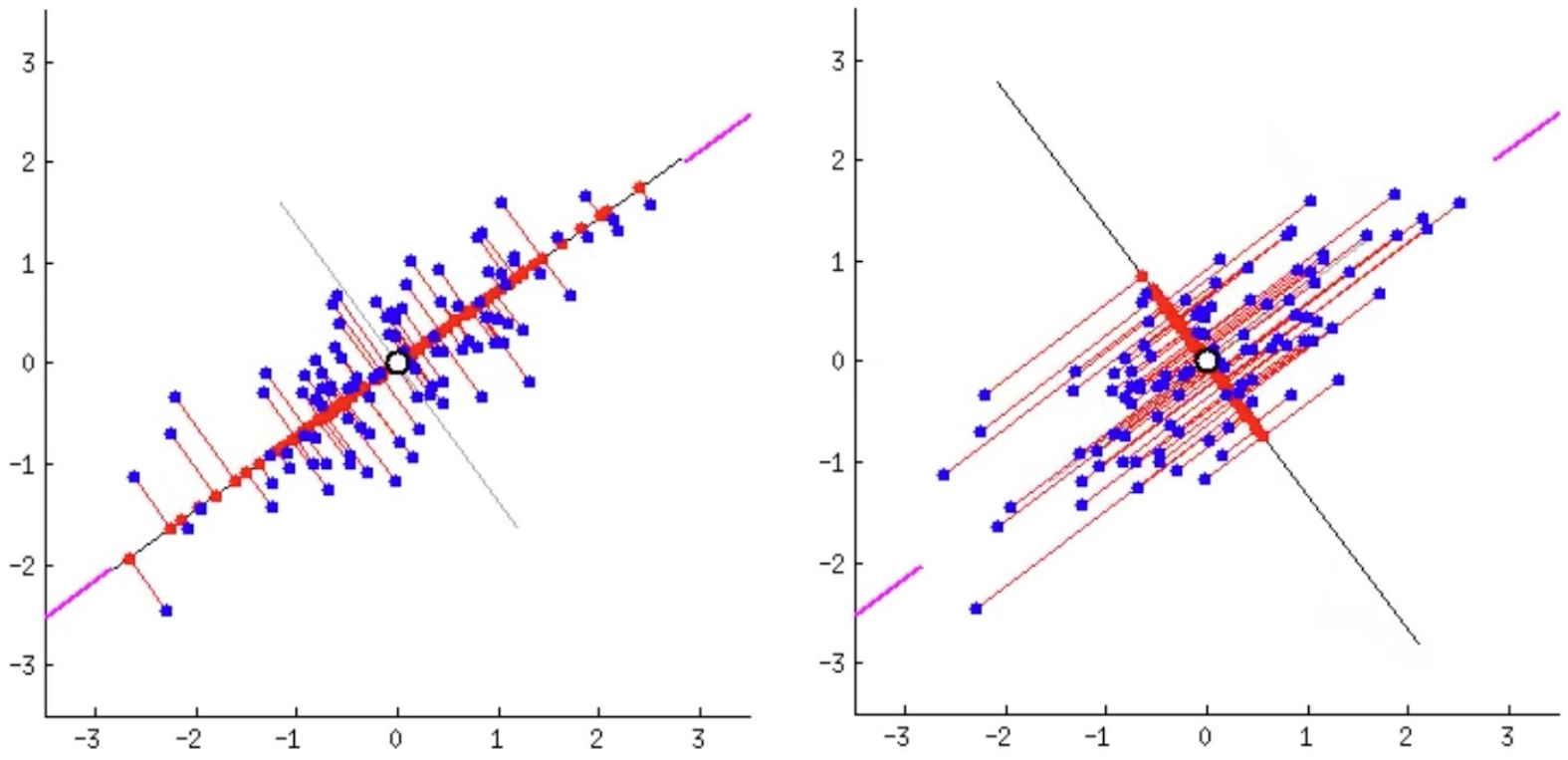
(b) Compressed dataset where only the x_1 coordinate is relevant.

One-dimensional PCA

One-dimensional PCA can reduces the high-dimensional data into a single score, which can then be used for

- Clustering
- Ranking
- Indexing

One-dimensional PCA



Maximizing the variance of the projection is equivalent to minimizing the average squared reconstruction error.

Derive one-dimensional PCA from First Principles

Problem Setting

- Assume the n-th observation is a D-dim vector

$$\boldsymbol{x}_n = \left[x_{n1}, x_{n2}, \cdots, x_{nD}\right]^\top$$

  and there are in total N observations.

- Stack each observation by row. The data matrix $\boldsymbol{X}$ is

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{x}_1^\top \\ \boldsymbol{x}_2^\top \\ \vdots \\ \boldsymbol{x}_N^\top \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1D} \\ x_{21} & x_{22} & \cdots & x_{2D} \\ \vdots & \vdots & & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{ND} \end{bmatrix}$$

Data Centering

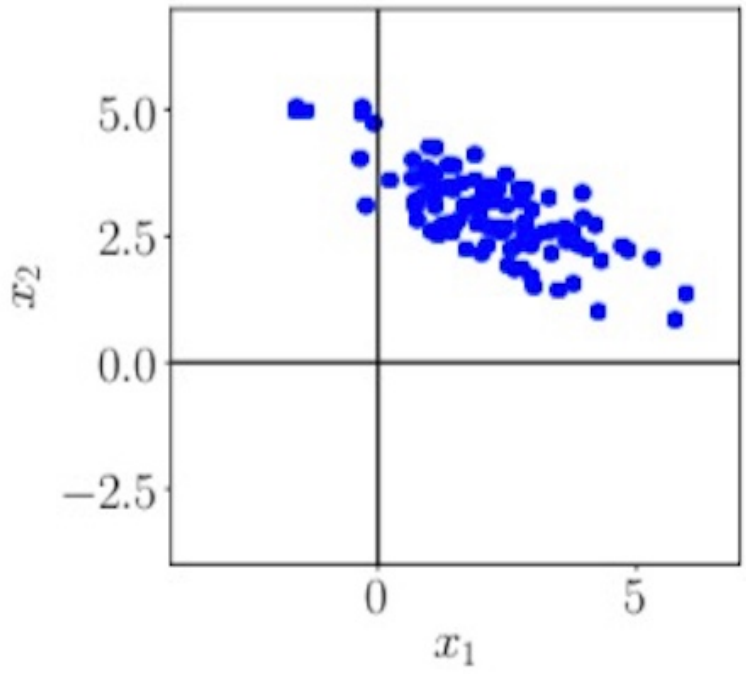
- For simplicity, we assume that the data is **centered** such that

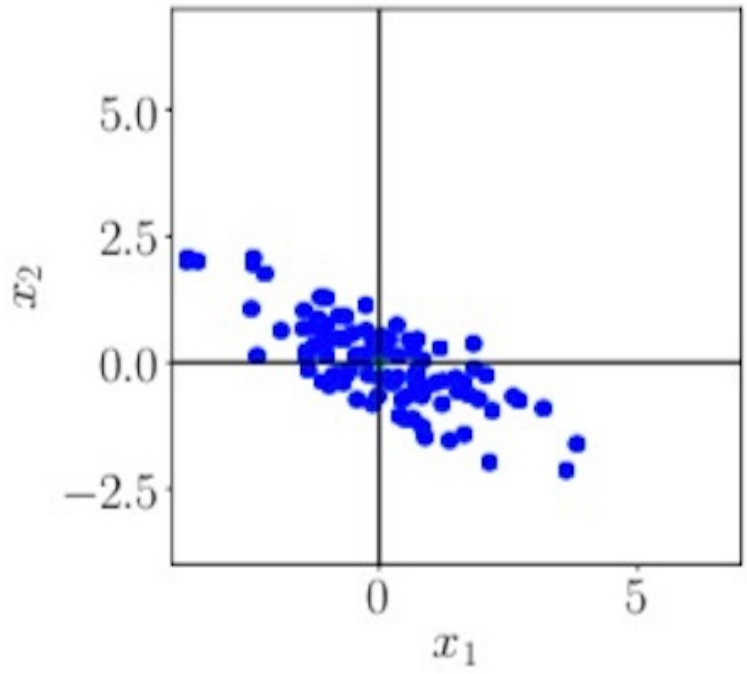
$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{x}_n = \mathbf{0}$$

- To achieve this, set

$$\boldsymbol{x}_n \leftarrow \boldsymbol{x}_n - \hat{\boldsymbol{\mu}}$$

Data Centering



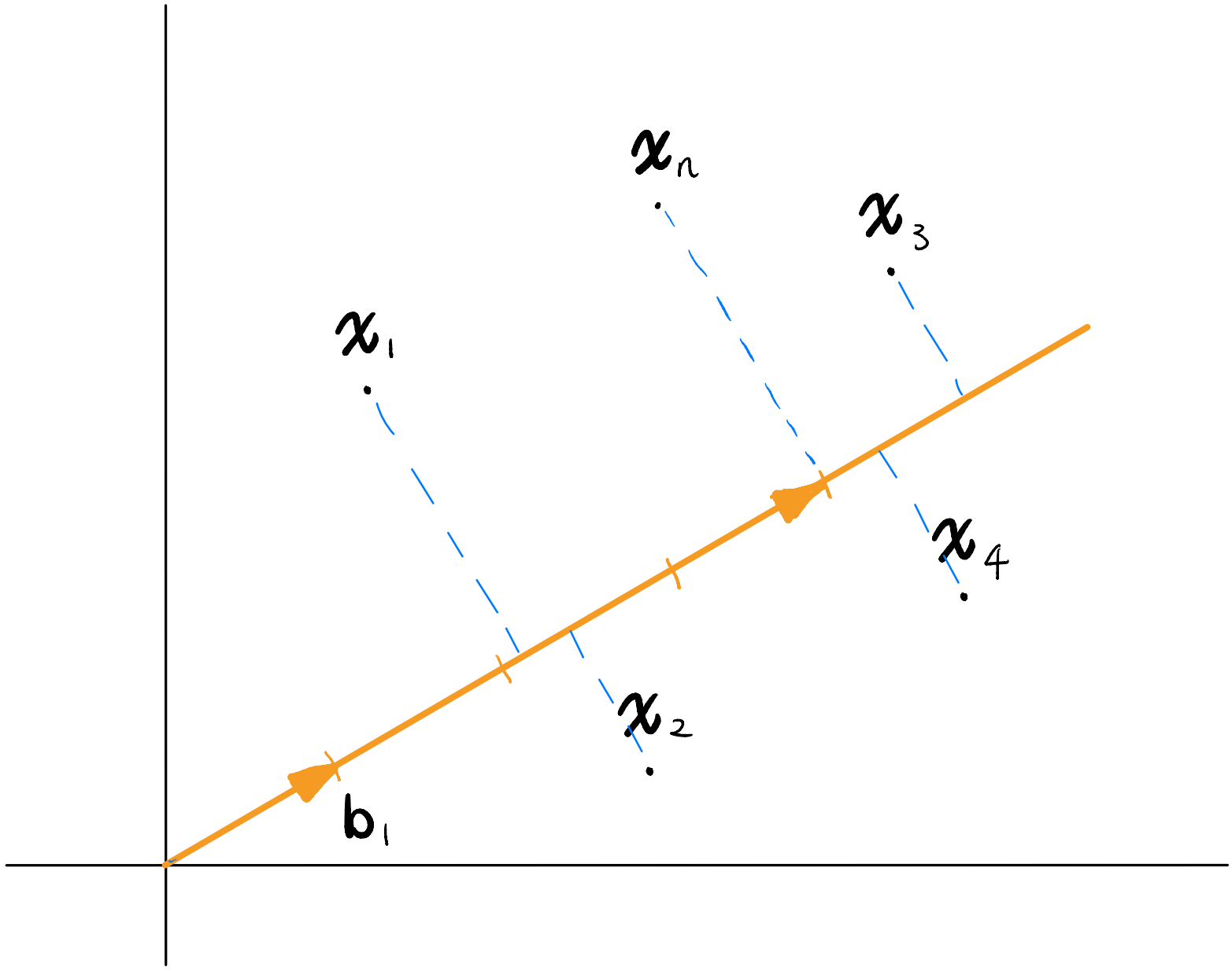


Data Centering

- **Centering** substracting each entry by the corresponding column mean:

$$\begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \cdots & x_{1D} - \bar{x}_D \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \cdots & x_{2D} - \bar{x}_D \\ \vdots & \vdots & & \vdots \\ x_{N1} - \bar{x}_1 & x_{N2} - \bar{x}_2 & \cdots & x_{ND} - \bar{x}_D \end{bmatrix}$$

Projection Perspective: One-dimensional Projection

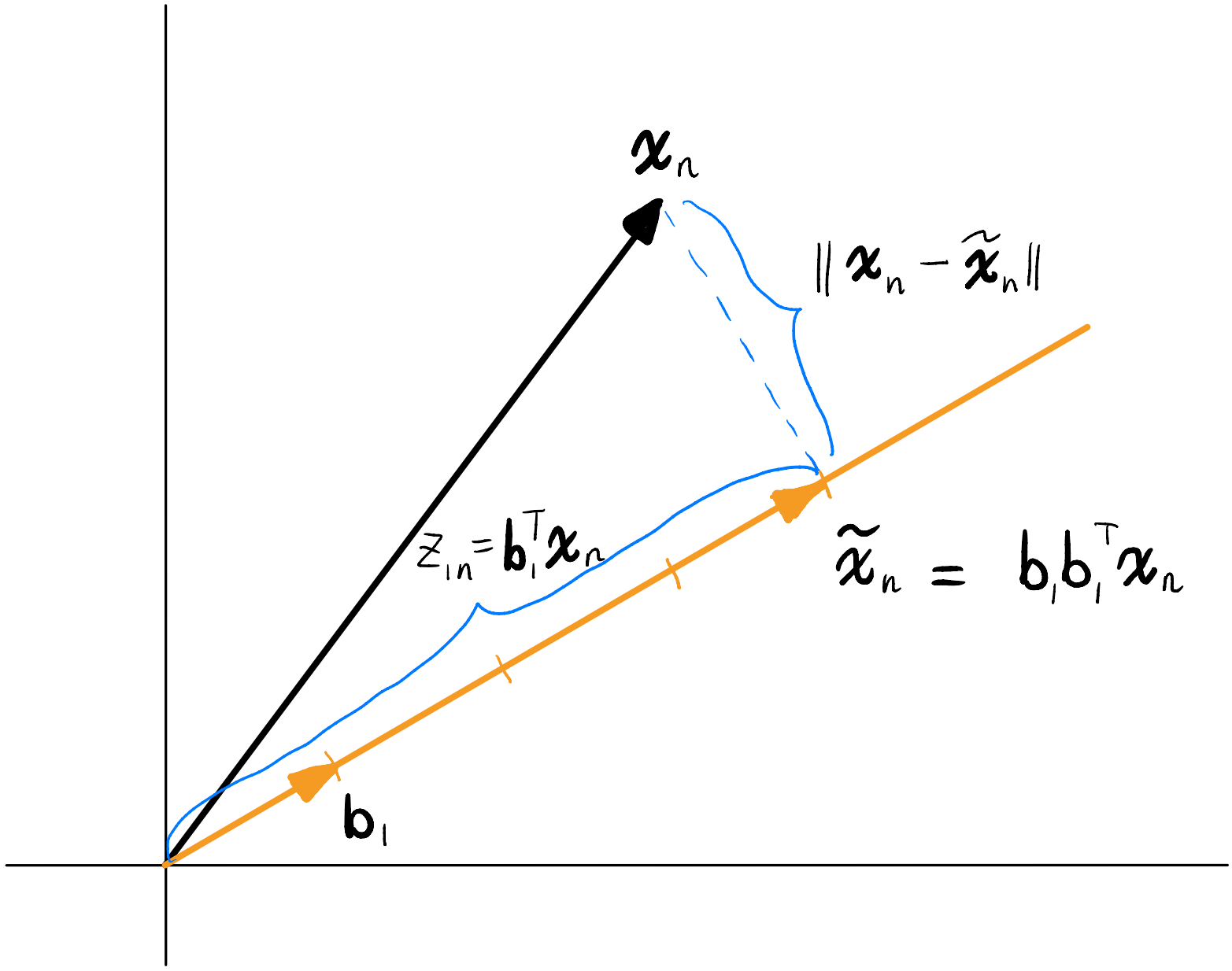

x_n
x_3
x_1
x_4
x_2
b_1

Minimizing reconstruction error

- Find projections $\tilde{\boldsymbol{x}}_n$ of data points $\boldsymbol{x}_n$
  - Similar to the original data points as possible
  - But have a significantly lower intrinsic dimensionality.
- We'll start by looking for a one-dimensional projection.
- Project N points $\boldsymbol{x}_n \in \mathbb{R}^D$, $n = 1, \cdots, N$ on to a line through the origin.
- We specify the line by a **unit column vector**

  $\boldsymbol{b}_1 = (b_{11}, \cdots, b_{1D})^\top \in \mathbb{R}^D$, with $\|\boldsymbol{b}_1\| = 1$, which forms a subspace U

$$U = \{z_1 \boldsymbol{b}_1 : \forall z_1 \in \mathbb{R}\}$$


x_n
$\|x_n - \tilde{x}_n\|$
$z_{1n} = b_1^T x_n$
$\tilde{x}_n = b_1 b_1^T x_n$
b_1

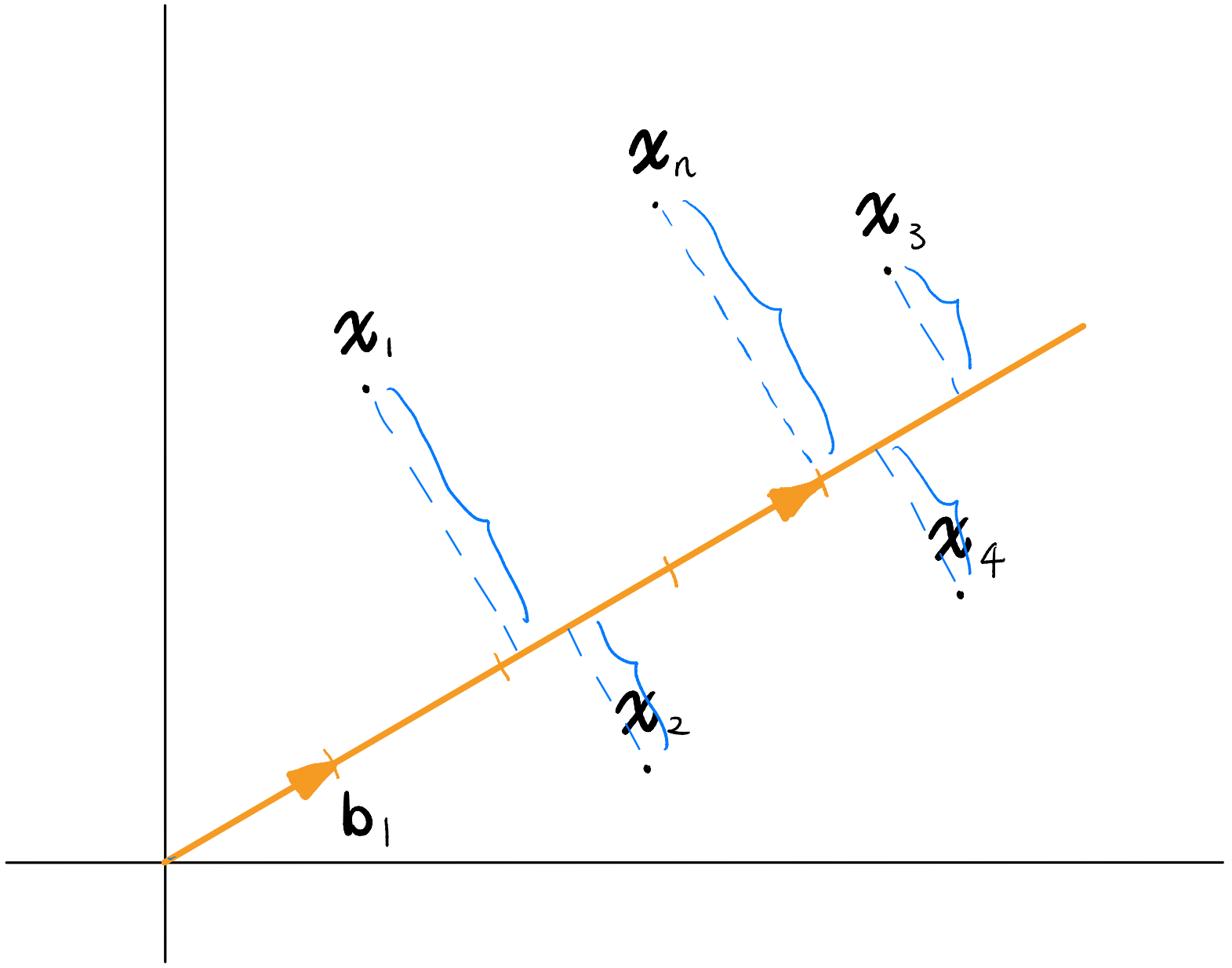
Minimizing reconstruction error

- The **coordinate** of the orthogonal projection of $\boldsymbol{x}_n$ onto the line U (one-dimensional subspace spanned by $\boldsymbol{b}_1$) is

$$z_{1n} = \boldsymbol{b}_1^\top \boldsymbol{x}_n$$

- The **orthogonal projection** $\tilde{\boldsymbol{x}}_n$ is

$$\tilde{\boldsymbol{x}}_n \equiv \pi_U(\boldsymbol{x}_n) = \underbrace{\boldsymbol{b}_1}_{\text{direction of proj.}} \cdot \underbrace{\boldsymbol{b}_1^\top \boldsymbol{x}_n}_{z_{1n}:\text{coordinate of proj.}}$$


x_n
x_3
x_1
x_4
x_2
b_1

Minimizing reconstruction error

- For n-th obs., approximate $\boldsymbol{x}_n$ using $\tilde{\boldsymbol{x}}_n$, the **squared approximation error** is

$$\|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2 = \|\boldsymbol{x}_n - \pi_U(\boldsymbol{x}_n)\|^2 = \|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n\|^2$$

- If average those $\|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2$ over all points $n = 1, \cdots, N$, the average squared approximation error, i.e. **reconstruction error** is

$$J_1 \equiv \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2 = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n\|^2 \quad \text{(reconstruction err.)}$$

Minimizing reconstruction error

- How to choose a good subspace U in PCA? – find $\boldsymbol{b}_1$ such that the reconstruction error is minimized.
- Optimization problem of 1-D PCA:

$$\min_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1} J_1 \iff \min_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1} \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n\right\|^2$$

Yin and Yang 阴阳 – data's duality

- Here is an amazing result that bridges together two perspectives of PCA

reconstruction err. const. projected variance

$$\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n - \underbrace{\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n}_{\tilde{\boldsymbol{x}}_n}\|^2 = \frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n\|^2 - \frac{1}{N}\sum_{n=1}^{N}\|\underbrace{\boldsymbol{b}_1^\top\boldsymbol{x}_n}_{z_{1n}}\|^2$$

- Minimizing the reconstruction error is equivalent to maximizing the variance of the projection.

$$\min_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1}\underbrace{\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n\|^2}_{J_1} \iff \max_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1}\underbrace{\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{b}_1^\top\boldsymbol{x}_n\|^2}_{V_1}$$

Proof

- The squared approximation error for point $\boldsymbol{x}_n$ can be rewritten as

$$
\begin{aligned}
\left\|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n\right\|^2 &= \left(\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n\right)^\top\left(\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n\right) \\
&= \boldsymbol{x}_n^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n + \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n \\
&\overset{(i)}{=} \boldsymbol{x}_n^\top\boldsymbol{x}_n - \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n \\
&= \left\|\boldsymbol{x}_n\right\|^2 - \left\|\boldsymbol{b}_1^\top\boldsymbol{x}_n\right\|^2 \\
&= \text{const} - z_{1n}^2
\end{aligned}
$$

where (i) follows by $\boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n \overset{\boldsymbol{b}_1^\top\boldsymbol{b}_1=1}{=} \boldsymbol{x}_n^\top\boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n$ and (ii) follows by $z_{1n} = \boldsymbol{b}_1^\top\boldsymbol{x}_n$.

V_1: sample variance of projected coordinates

- (Homework) Show that the term $V_1 = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{b}_1^\top \boldsymbol{x}_n\|^2$ indeed is equal to the **sample variance of the coodinates** $\{z_{11}, \cdots, z_{1N}\}$ of N projections.

$$V_1 = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{b}_1^\top \boldsymbol{x}_n\|^2 = \frac{1}{N}\sum_{n=1}^{N} z_{1n}^2$$
$$= \frac{1}{N}\sum_{n=1}^{N}\left(z_{1n} - \frac{1}{N}\sum_{n=1}^{N} z_{1n}\right)^2 \equiv \widehat{\mathrm{Var}}(z_{11}, \cdots, z_{1N})$$

- **Hint:** recall that the data is centered; also note that we use factor $\frac{1}{N}$ instead of $\frac{1}{N-1}$ in the sample variance.
- Note that $\widehat{\mathrm{Var}}$ stands for the sample variance, rather than the population variance Var.

- (Homework) Show that we also have

$$\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2 = \frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n\|^2 - \frac{1}{N}\sum_{n=1}^{N}\|\tilde{\boldsymbol{x}}_n\|^2$$

  i.e. norm of reconstruction is equal to norm of coordinate. (If you draw it, this is obvious).

Maximum Variance Perspective

- We can further show that

$$\min_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1} \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top \boldsymbol{x}_n\|^2 \iff \max_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1} \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{b}_1^\top \boldsymbol{x}_n\|^2$$

$$\iff \max_{\boldsymbol{b}_1:\|\boldsymbol{b}_1\|=1} \boldsymbol{b}_1^\top \boldsymbol{S}\boldsymbol{b}_1$$

where $\boldsymbol{S}$ is the **sample (data) covariance matrix**

$$\boldsymbol{S} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n\boldsymbol{x}_n^\top = \frac{1}{N}\boldsymbol{X}^\top\boldsymbol{X}$$

Proof

- For simplicity, rewrite the problem in a matrix format

$$
\begin{aligned}
V_1 &= \frac{1}{N}\sum_{n=1}^{N} \left\| \boldsymbol{b}_1^\top \boldsymbol{x}_n \right\|^2 \\
&= \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{b}_1^\top \boldsymbol{x}_n \boldsymbol{x}_n^\top \boldsymbol{b}_1 = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{b}_1^\top \boldsymbol{x}_n \boldsymbol{x}_n^\top \boldsymbol{b}_1 \\
&= \boldsymbol{b}_1^\top \left(\frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^\top \right) \boldsymbol{b}_1 \\
&= \boldsymbol{b}_1^\top \left(\frac{1}{N} \boldsymbol{X}^\top \boldsymbol{X} \right) \boldsymbol{b}_1 \\
&= \boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1
\end{aligned}
$$

Direction with maximal variance

We will show that the optimization problem

$$\max_{\boldsymbol{b}_1 : \|\boldsymbol{b}_1\|=1} \boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1$$

reaches its maximal value λ_1 at maximizer $\boldsymbol{b}_1$

$$\lambda_1 = \max_{\|\boldsymbol{b}\|=1} \boldsymbol{b}^\top \boldsymbol{S} \boldsymbol{b}, \qquad \boldsymbol{b}_1 = \arg\max_{\|\boldsymbol{b}\|=1} \boldsymbol{b}^\top \boldsymbol{S} \boldsymbol{b}$$

where λ_1 is the largest eigenvalue of $\boldsymbol{S}$ and $\boldsymbol{b}_1$ is the corresponding eigenvector.

Direction with maximal variance

- To solve[1]

$$\max_{\boldsymbol{b}_1} \boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1$$

$$\text{subject to } \|\boldsymbol{b}_1\| = 1$$

Introduce the **Lagrangian function** $\mathcal{L}$

$$\mathcal{L}(\boldsymbol{b}_1, \lambda_1) = \boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1 + \lambda_1 (1 - \boldsymbol{b}_1^\top \boldsymbol{b}_1)$$

[1] Note that arbitrarily increasing the magnitude of the vector $\boldsymbol{b}_1$ increases $\boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1$. Thus we have to restrict $\boldsymbol{b}_1$ to be **a unit vector**, i.e. $\|\boldsymbol{b}_1\| = 1$

Direction with maximal variance

- The partial derivatives of $\mathcal{L}$ wrt $\boldsymbol{b}_1$ and λ_1 are

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}_1} = 2\boldsymbol{b}_1^\top \boldsymbol{S} - 2\lambda_1 \boldsymbol{b}_1^\top \qquad \frac{\partial \mathcal{L}}{\partial \lambda_1} = 1 - \boldsymbol{b}_1^\top \boldsymbol{b}_1$$

- Setting these partial derivatives to $\mathbf{0}$

$$2\boldsymbol{b}_1^\top \boldsymbol{S} - 2\lambda_1 \boldsymbol{b}_1^\top = \mathbf{0} \qquad \frac{\partial \mathcal{L}}{\partial \lambda_1} = 1 - \boldsymbol{b}_1^\top \boldsymbol{b}_1 = 0$$

  which gives

$$\boldsymbol{S}\boldsymbol{b}_1 = \lambda_1 \boldsymbol{b}_1 \qquad \boldsymbol{b}_1^\top \boldsymbol{b}_1 = 1$$

- Lagrange multiplier λ_1 is the **eigenvalue**
- **1st Principal Component:** $\boldsymbol{b}_1$ is the corresponding **eigenvector** of matrix $\boldsymbol{S}$.

- In this case, the variance of the projected coordinates is

$$V_1 = \frac{1}{N}\sum_{n=1}^{N} \left\| \boldsymbol{b}_1^\top \boldsymbol{x}_n \right\|^2 = \boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1 = \boldsymbol{b}_1^\top \lambda_1 \boldsymbol{b}_1 = \lambda_1$$

- To maximize V_1, choose λ_1 to be the **largest eigenvalue** of $\boldsymbol{S}$, and $\boldsymbol{b}_1$ is the corresponding **eigenvector.**

- Total variance explained by projection

$$\frac{\frac{1}{N}\sum_{n=1}^{N} \left\| \boldsymbol{b}_1^\top \boldsymbol{x}_n \right\|^2}{\frac{1}{N}\sum_{n=1}^{N} \left\| \boldsymbol{x}_n \right\|^2} = \frac{\lambda_1}{\frac{1}{N}\sum_{n=1}^{N} \left\| \boldsymbol{x}_n \right\|^2}$$

1D PCA in Practice

Tokyo Medal Ranking

- 2020 Summer Olympics, Tokyo; 93 nations received at least one medal. How do we rank them?

Country	Gold	Silver	Bronze
US	39	41	33
China	38	32	19
Japan	27	14	17
UK	22	20	22
ROC	20	28	23
Australia	17	7	22
Netherlands	10	12	14
France	10	12	11
...			

Which you choose?

- **Gold First Method** – standard convention used by the International Olympic Committee
- **Medal Count Method** – primarily used by US (and occasionally Canada)

Let data make the call

- Apply 1D PCA on X = Data:

$X =$

Country	Gold	Silver	Bronze
US	39	41	33
China	38	32	19
Japan	27	14	17
UK	22	20	22
ROC	20	28	23
Australia	17	7	22
Netherlands	10	12	14
France	10	12	11
...			

Let data make the call

- We find

$$\boldsymbol{b}_1 = \left[-0.581, -0.581, -0.568\right]^\top$$

- For any country n, the vector of medal counts $\boldsymbol{x}_n$ is

$$\boldsymbol{x}_n = \left[\text{gold}_n, \text{silver}_n, \text{bronze}_n\right]^\top$$

Let data make the call

- Project $\boldsymbol{x}_n$ onto $\boldsymbol{b}_1$ to obtain the coordinate of the projected data

$$z_{1n} = \boldsymbol{b}_1^\top \boldsymbol{x}_n = -0.581 \times \text{gold}_n \; -0.581 \times \text{silver}_n \; -0.568 \times \text{bronze}_n$$

  for $n = 1, 2, 3$ (US, China, Japan...)

- Notice that the weights are quite similar for gold, silver and bronze medals, suggesting **Medal Count Method**

- Total variance explained by projection is 72%

$$\frac{\lambda_1}{\frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n\|^2} = \frac{2.74}{2.96} = 0.92$$

  suggesting good approximation.

Interpretation of PCA Results

Interpretation

1. **Eigenvector $b_1 \in \mathbb{R}^D$: 1st principal component.**

   Primary axis of variation – the direction in the data that captures the greatest amount of variance.

2. **Eigenvalue λ_1:** variance V_1 explained by the 1st PC.

3. **PC scores:** the projection coordinates $z_{11}, \cdots, z_{1N}$

$$\boldsymbol{X}\boldsymbol{b}_1 = \begin{bmatrix} \boldsymbol{b}_1^\top \boldsymbol{x}_1 \\ \boldsymbol{b}_1^\top \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{b}_1^\top \boldsymbol{x}_N \end{bmatrix} = \begin{bmatrix} z_{11} \\ z_{12} \\ \vdots \\ z_{1N} \end{bmatrix}$$

   represent one-dimensional embeding of the original N data.

Loadings of PC

- **Loadings of a PC:** $b_{11}, b_{12}, \cdots, b_{1D}$ are the elements of $\boldsymbol{b}_1$

$$\boldsymbol{b}_1 = (b_{11}, b_{12}, \cdots, b_{1D})^\top \in \mathbb{R}^D$$

- The k-th element, b_{1k} indicate the contribution (or weight) of the k-th original variable to the first PC, for $k = 1, \cdots, D$.

Loadings of PC

- Let $\boldsymbol{z}_1$ be the vector of PC scores for observation $n = 1, \cdots, N$ and X_k be N observations of k-th original variable

$$\begin{bmatrix} \overbrace{\begin{matrix} z_{11} \\ z_{12} \\ \vdots \\ z_{1N} \end{matrix}}^{\boldsymbol{z}_1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}_1^\top \boldsymbol{x}_1 \\ \boldsymbol{b}_1^\top \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{b}_1^\top \boldsymbol{x}_N \end{bmatrix} = \boldsymbol{X}\boldsymbol{b}_1 \qquad \boldsymbol{X} = \begin{bmatrix} x_{11} \cdots & \overbrace{\begin{matrix} x_{1k} \\ x_{2k} \\ \vdots \\ x_{Nk} \end{matrix}}^{X_k} & \cdots x_{1D} \\ x_{21} \cdots & & \cdots x_{2D} \\ \vdots & & \vdots \\ x_{N1} \cdots & & \cdots x_{ND} \end{bmatrix}$$

- We can show that

$$(b_{11}, b_{12}, \cdots, b_{1D})^\top = \left[\frac{\widehat{\mathrm{Cov}}(X_1, \boldsymbol{z}_1)}{\lambda_1}, \frac{\widehat{\mathrm{Cov}}(X_2, \boldsymbol{z}_1)}{\lambda_1}, \cdots, \frac{\widehat{\mathrm{Cov}}(X_D, \boldsymbol{z}_1)}{\lambda_1} \right]^\top$$

Proof

- Recall the definition of sample covariance between two vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^N$

$$\widehat{\mathrm{Cov}}(\boldsymbol{a}, \boldsymbol{b}) = \frac{1}{N} \boldsymbol{a}^\top \boldsymbol{b} = \frac{1}{N} \sum_{n=1}^{N} a_n b_n$$

  if both vectors are assumed to have mean zero.

- Thus, the sample covariance between X_k and $\boldsymbol{z}_1$ is

$$\widehat{\mathrm{Cov}}(X_k, \boldsymbol{z}_1) = \frac{1}{N} X_k^\top \boldsymbol{z}_1 = \frac{1}{N} X_k^\top (\boldsymbol{X} \boldsymbol{b}_1) = \left[\boldsymbol{S} \boldsymbol{b}_1\right]_k = \left[\lambda_1 \boldsymbol{b}_1\right]_k = \lambda_1 b_{1k}$$

- Therefore the loadings are

$$(b_{11}, b_{12}, \cdots, b_{1D})^\top = \left[\frac{\widehat{\mathrm{Cov}}(X_1, \boldsymbol{z}_1)}{\lambda_1}, \frac{\widehat{\mathrm{Cov}}(X_2, \boldsymbol{z}_1)}{\lambda_1}, \cdots, \frac{\widehat{\mathrm{Cov}}(X_D, \boldsymbol{z}_1)}{\lambda_1} \right]^\top$$

Interpretation

- **Reconstructed data:** $\tilde{\boldsymbol{x}}_1, \cdots, \tilde{\boldsymbol{x}}_N$ are reconstruction (approximation) of $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N$ by mapping the coordinate z_{1n} back into data space

$$\tilde{\boldsymbol{x}}_n = \boldsymbol{b}_1 z_{1n} = \boldsymbol{b}_1 \boldsymbol{b}_1^\top \boldsymbol{x}_n \in \mathbb{R}^D$$

  for $n = 1, \cdots, N$

- Although $\tilde{\boldsymbol{x}}_n$ is a D-dimensional vector, it only requires a single coordinate z_{1n} to represent it with respect to the basis vector $\boldsymbol{b}_1 \in \mathbb{R}^D$.

Reconstruction error

If we project the data to the first principal component $\boldsymbol{b}_1$, we see that the corresponding reconstruction error is

$$
\begin{aligned}
J_1 &\equiv \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\right\|^2 \\
&= \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{x}_n\right\|^2 - \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{b}_1^\top \boldsymbol{x}_n\right\|^2 \\
&= \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{x}_n\right\|^2 - \boldsymbol{b}_1^\top \boldsymbol{S} \boldsymbol{b}_1 \\
&= \frac{1}{N}\sum_{n=1}^{N} \left\|\boldsymbol{x}_n\right\|^2 - \underbrace{\lambda_1}_{\text{variance explained by PC1}}
\end{aligned}
$$

Data Centering and Scaling

Data centering and scaling

- In PCA, the **centering** and **scaling** of data are important steps.
- But their necessity depends on the context and nature of your data.
- The original data matrix is

$$\boldsymbol{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1D} \\ x_{21} & x_{22} & \cdots & x_{2D} \\ \vdots & \vdots & & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{ND} \end{bmatrix}$$

- **Centering** is almost always necessary in PCA.
- Involves substracting each entry by the corresponding column mean:

$$\begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \cdots & x_{1D} - \bar{x}_D \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \cdots & x_{2D} - \bar{x}_D \\ \vdots & \vdots & & \vdots \\ x_{N1} - \bar{x}_1 & x_{N2} - \bar{x}_2 & \cdots & x_{ND} - \bar{x}_D \end{bmatrix}$$

- The mean of each column **after centering** becomes zero.
- Ensures that each variable contributes equally to the analysis so that PCA focuses on the variance of the data.
- Without, PCA might be influenced by variables that are on a larger scale.

- **Scaling** - aka **standardization**
- Involves dividing each entry by the corresponding column standard deviation:

$$\begin{bmatrix} \frac{x_{11}-\bar{x}_1}{\sqrt{\widehat{\mathrm{Var}}(x_{11},\cdots,x_{N1})}} & \frac{x_{12}-\bar{x}_2}{\sqrt{\widehat{\mathrm{Var}}(x_{12},\cdots,x_{N2})}} & \cdots & \frac{x_{1D}-\bar{x}_D}{\sqrt{\widehat{\mathrm{Var}}(x_{1D},\cdots,x_{ND})}} \\ \frac{x_{21}-\bar{x}_1}{\sqrt{\widehat{\mathrm{Var}}(x_{11},\cdots,x_{N1})}} & \frac{x_{22}-\bar{x}_2}{\sqrt{\widehat{\mathrm{Var}}(x_{12},\cdots,x_{N2})}} & \cdots & \frac{x_{2D}-\bar{x}_D}{\sqrt{\widehat{\mathrm{Var}}(x_{1D},\cdots,x_{ND})}} \\ \vdots & \vdots & & \vdots \\ \frac{x_{N1}-\bar{x}_1}{\sqrt{\widehat{\mathrm{Var}}(x_{11},\cdots,x_{N1})}} & \frac{x_{N2}-\bar{x}_2}{\sqrt{\widehat{\mathrm{Var}}(x_{12},\cdots,x_{N2})}} & \cdots & \frac{x_{ND}-\bar{x}_D}{\sqrt{\widehat{\mathrm{Var}}(x_{1D},\cdots,x_{ND})}} \end{bmatrix}$$

- The column variance of each column **after centering** becomes one.
- Crucial when the variables in data are **on different scales** (e.g., kg, km, F, C) or have different units of measurement.
- Scaling ensures that PCA gives equal weight to each variable, preventing variables with larger scales from dominating the PCA.

- Always center the data;
- Scale the data when variables are on different scales or units;
  - In imaging — where all coordinates are in the same units, namely pixel intensities — there is no need to do such coordinate scaling.
- Be cautious with scaling for interpretability.

Summary of PCA

1. Mean substraction (centering) and scaling

$$x_{nd} \leftarrow \frac{x_{nd} - \bar{x}_d}{\mathrm{sd}_d}$$

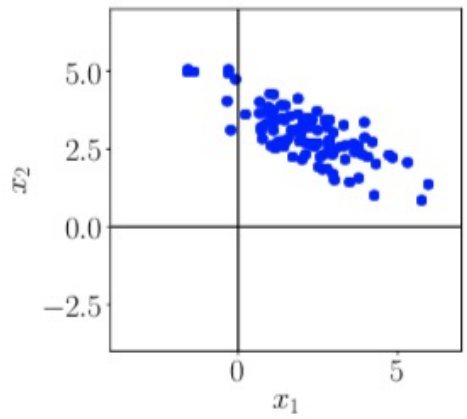
where

$$\bar{x}_d = \frac{1}{N} \sum_{n=1}^{N} x_{nd}$$

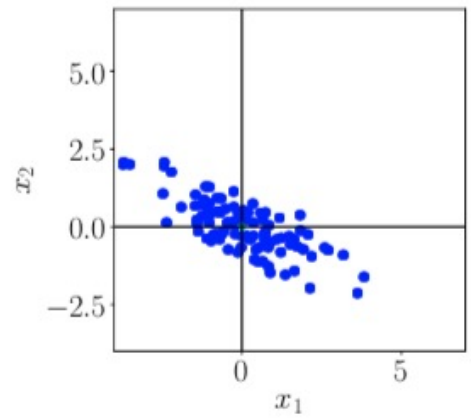
$$\mathrm{sd}_d \equiv \sqrt{\widehat{\mathrm{Var}}(x_{1d}, \cdots, x_{Nd})} = \sqrt{\frac{1}{N-1} \sum_{n=1}^{N} (x_{nd} - \bar{x}_d)^2}$$

2. Eigendecomposition of the covariance matrix $\boldsymbol{S} = \frac{1}{N} \boldsymbol{X}^\top \boldsymbol{X}$
3. Projection $\tilde{\boldsymbol{x}}_n = \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}$ with coordinates $\boldsymbol{z}_n = \boldsymbol{B}^\top \boldsymbol{x}_n \in \mathbb{R}^M$
4. Undo centering and scaling of $\tilde{\boldsymbol{x}}_n = (\tilde{x}_{n1}, \cdots, \tilde{x}_{nD})^\top \in \mathbb{R}^D$. Let $\tilde{x}_{nd}$ be the d-th entry of $\tilde{\boldsymbol{x}}_n$.

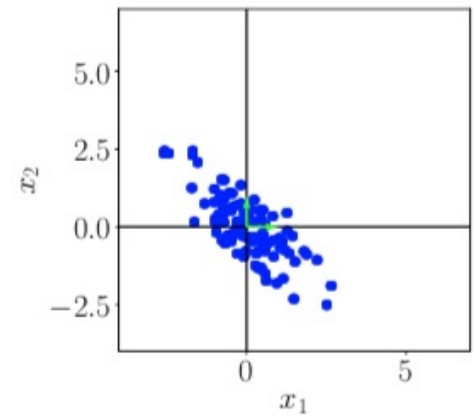
$$\tilde{x}_{nd} \leftarrow \tilde{x}_{nd} \mathrm{sd}_d + \bar{x}_d$$



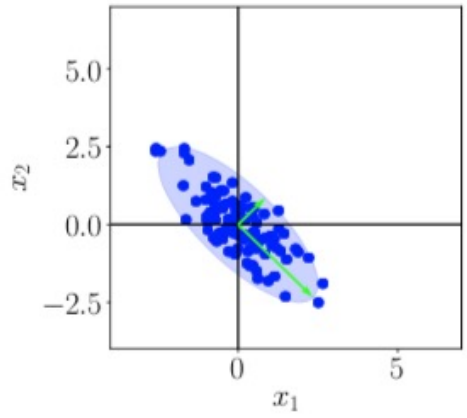
(a) Original dataset.



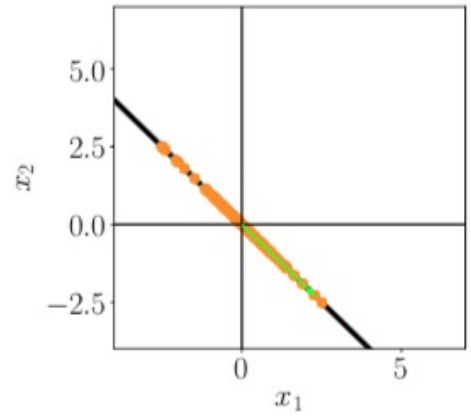
(b) Step 1: Centering by subtracting the mean from each data point.



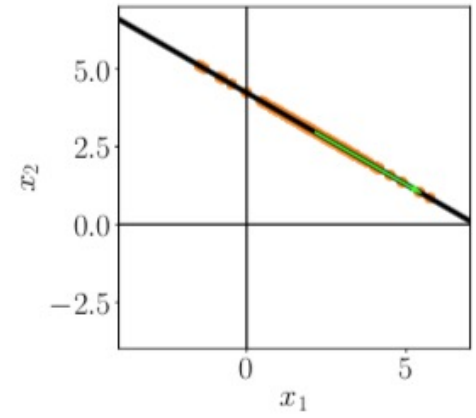
(c) Step 2: Dividing by the standard deviation to make the data unit free. Data has variance 1 along each axis.



(d) Step 3: Compute eigenvalues and eigenvectors (arrows) of the data covariance matrix (ellipse).



(e) Step 4: Project data onto the principal subspace.



(f) Undo the standardization and move projected data back into the original data space from (a).

Why use Orthogonal Projection in Reconstruction?

Why orthogonal projection is optimal?

- Recall that **orthogonal** projection is adopted to get reconstruction $\tilde{\boldsymbol{x}}_n$ in

$$J_1 \equiv \frac{1}{N}\sum_{n=1}^{N}\left\|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\right\|^2 = \frac{1}{N}\sum_{n=1}^{N}\left\|\boldsymbol{x}_n - \boldsymbol{b}_1\boldsymbol{b}_1^\top\boldsymbol{x}_n\right\|^2$$

- But **why** use orthogonal projection?
- Consider an unknown linear projection method, the resulting coordinate of the projection for data point $\boldsymbol{x}_n$ is z_{1n}
- The corresponding reconstruction error is

$$J_1 = \frac{1}{N}\sum_{n=1}^{N}\left\|\boldsymbol{x}_n - \boldsymbol{b}_1 z_{1n}\right\|^2$$

- Assume $\boldsymbol{b}_1$ is given, find z_{1n} that minimize the reconstruction error

$$\min_{z_{11},\cdots,z_{1N}} \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \boldsymbol{b}_1 z_{1n}\|^2$$

- To solve this problem, compute the partial derivative and set it to zero

$$\frac{\partial J_1}{\partial z_{1n}} = -\frac{2}{N}(\boldsymbol{x}_n - \boldsymbol{b}_1 z_{1n})^\top \boldsymbol{b}_1 = -\frac{2}{N}(\boldsymbol{b}_1^\top \boldsymbol{x}_n - \boldsymbol{b}_1^\top \boldsymbol{b}_1 z_{1n}) = 0$$

- Since $\boldsymbol{b}_1^\top \boldsymbol{b}_1 = 1$, the equation yields

$$z_{1n} = \boldsymbol{b}_1^\top \boldsymbol{x}_n$$

- Consequently,
  - The optimal linear projection $\tilde{\boldsymbol{x}}_n$ of $\boldsymbol{x}_n$ is an orthogonal projection.
  - **Optimal** coordinates z_{1n} of the projection $\tilde{\boldsymbol{x}}_n$ are the coordinates of the **orthogonal projection** of the original data point $\boldsymbol{x}_n$.